MIL-C-24643/18B
3 October 1986
SUPERSEDING
MIL-C-0024643/18A(SH)
8 November 1985
MIL-C-24643/18
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 1000 VOLTS, TYPE LSMSCU (INCLUDING VARIATIONS LSMSCA AND LSMSCS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

This cable can be manufactured in three variations: LSMSCU (unarmored), LSMSCA (armored) and LSMSCS (double overall shield).

REQUIREMENTS:

Qualification required.

Construction (watertight with circuit integrity)

First - Copper conductor, uncoated, AWG size 18, class B stranding.
Second - Extruded silicone rubber insulation. Nominal thickness of 0.018 inch. Diameter over insulation 0.084-inch minimum.
Third - Glass braid.
Fourth - Braid covering. Standard identification code applied by method 1. Diameter over braid covering 0.113 inch maximum.
Fifth - The required number of conductors (see table I) cabled together with a lay not greater than 24 times the pitch diameter of the layer, except the lay of the two outer layers for sizes 44, 61 and 91 shall not exceed 18 times the pitch diameter of the layer. Fillers shall be used to form a firm, well-rounded assembly. Cabling sequence shall be consecutive, starting with no. 1, from the center outward.
Sixth - Binder or combination binder and barrier.

(B) denotes changes.

AMSC N/A FSC 6145
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Construction variant for type LSMSCU

Seventh - Cross-linked polyolefin jacket (see table I for thickness). Cable surface marking required.

Construction variant for type LSMSCA

Seventh - Cross-linked polyolefin jacket (see table I for thickness).
Eighth - Braided metal armor and paint.

Construction variant for type LSMSCS

Seventh - Braided shield of AWG no. 34 tin coated copper. Braid angle of 20 to 40 degrees; minimum coverage of 85 percent.
Eighth - Spirally wrapped 0.002-inch thick polyethylene terephthalate (mylar) separator tape with 25 percent minimum overlap.
Ninth - Braided shield of AWG no. 34 tin coated copper. Braid angle of 20 to 40 degrees; minimum coverage of 85 percent.
Tenth - Binder tape applied helically with overlap.
Eleventh - Cross-linked polyolefin jacket (see table I for thickness). Cable surface marking required.

TABLE I. Details.

Military 1/ part no. M24643/18	Size 2/	Number of conductors	Jacket thickness nominal, inch	Overall diameter (inches)					
				Type LSMSCU		Type LSMSCA		Type LSMSCS	
				min	max	min	max	min	max
-01UN	7	7	0.040	0.447	0.484	0.497	0.534	0.507	0.544
-02UN	10	10	.050	.575	.622	.625	.672	.635	.682
-03UN	14	14	.050	.617	.668	.667	.718	.677	.728
-04UN	19	19	.050	.682	.738	.732	.788	.742	.798
-05UN	24	24	.050	.790	.855	.840	.905	.850	.915
-06UN	30	30	.050	.833	.901	.883	.951	.893	.961
-07UN	37	37	.060	.926	1.002	.976	1.055	.986	1.065
-08UN	44	44	.060	1.030	1.114	1.080	1.164	1.090	1.174
-09UN	61	61	.060	1.156	1.250	1.206	1.300	1.216	1.310
-10UN	91	91	.060	1.369	1.480	1.419	1.530	1.429	1.540

1/ Unarmored. When armor option or double overall shield is required, see MIL-C-24643 for configuration of part number.
(B) 2/ Type LSMSCU, LSMSCA, and LSMSCS.

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EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	7.07
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor.....	3000
Insulation resistance - shield to shield	
(LSMSCS only, overall shields only), volts, minimum	6
Megohms/1000 feet minimum.....	1.0
Conductor to conductor.....	500
Jacket flaws.....	No failure
Conductor and shield (LSMSCS only) continuity.....	No failure
Group A:	
Visual and dimensional.....	No failure
Watertightness - see MIL-C-24643 for limits of water leakage.....	No failure
Abrasion resistance - scrapes, minimum.....	250
(B) Crack resistance.....	No damage
Group B:	
Gas flame[1/] - 1 hour.....	No failure
Drip - 95 ± 1°C.....	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)..	35
Cold bending, cable - maximum mandrel diameter 12 times cable diameter.....	No damage
Physicals (unaged)	
Insulation	
Tensile strength - lbf/in^2, minimum.....	750
Elongation - percent, minimum.....	125
Jacket (cable)	
Tensile strength - lbf/in^2, minimum.....	1300
Elongation - percent, minimum.....	160
Group C:	
Physicals (aged)	
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum.....	60
Elongation - percent of unaged, minimum..	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum.....	50
Elongation - percent, minimum.....	50
Shrinkage.....	No failure
Heat distortion - percent of unaged, maximum.....	30
Permanence of printing (conductor) - cycles, minimum....	50
Permanence of printing (jacket) - (types LSMSCU and LSMSCS) cycles, minimum.....	250

1/ Type LSMSCS only. Shields shall not be grounded during test.

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EXAMINATION AND TESTS: (Continued)

	Requirements
Cable filler removability	No failure
Armor (type LSMSCA only) - conformance to material, construction and coverage	No failure
Shield (type LSMSCS only) - conformance to material, construction and coverage	No failure
(B) Surface transfer impedance (type LSMSCS only)	
Milliohms per meter, maximum	700
EMP response time - ms/500 feet	60

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Cable aging and compatibility (125 ± 5°C)	No failure
Cold working (minus 20 ± 2°C)	No damage
Gas flame 1/ - (3 hours)	No failure
Specific gravity of extruded silicone insulation, maximum	1.55
Abrasion resistance (jacket)	No failure
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation (cable)	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	—
(B) Insulation	35
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

1/ Type LSMSCS only. Shield shall not be grounded during test.

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSMSCU-7, LSMSCA-7, LSMSCS-7	1500
LSMSCU-10, LSMSCA-10, LSMSCS-10	1500
All others	1000

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NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-1005-18)

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